Markovianity

We can keep only neighborhood of x_i and not all the x_j... $p(x_1) \prod_{i=2}^{N} p(x_i \mid x_{i-1})$

We can obtain some sofistical distribution by applying a succession of linear transformation with small non-linear transformation at each step...

Conditional independence

x_1 and x_3 are independent ACCORDING to x_2 given.

$p(x_1, x_2, x_3) = p(x_2)p(x_1 \mid x_2)p(x_3 \mid x_2)$

Unfolding: decompose very complex form to linear forms...

All vectors generated by a gaussian distribution have the same euclidean norm !

Taking L_1, L_2, L_3 etc... norms is the same than taking 1, 2, 3 moments

Le Jepa 12 Nov 2025 Yann Yelem

Batch norm: we must make all vectors having the same norm to right train the AI models !!!